

## SUMMERSIDE, PE, Canada

WMO: 717020

Lat: 46.4413N

Lon: 63.8375W

Elev: 12

StdP: 101.18

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -19.3      | -17.1 | -24.3                         | 0.4 | -18.8 | -21.7 | 0.5 | -16.4 | 16.5                  | -5.1 | 14.7 | -4.7 | 4.9                   | 270  | 0.699 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.0                    | 26.9            | 20.7 | 25.5 | 19.7 | 24.2 | 19.0 | 22.0                | 25.5 | 21.0 | 24.0 | 20.2 | 22.9 | 5.4                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.7                            | 15.4 | 24.0 | 19.9 | 14.6 | 23.0 | 19.1 | 13.9 | 22.1 | 64.4          | 25.6 | 61.0 | 24.0 | 58.0 | 22.9 | 25.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 13.4          | 11.7 | 10.4 | DB                         | -22.9 | 30.2               | 2.5 | 1.5                                                | -24.7 | 31.2       | -26.2 | 32.1       | -27.6 | 33.0       | -29.4 | 34.1 | (4) |
| (5)               |      |      | WB                         | -23.4 | 23.7               | 2.4 | 0.8                                                | -25.1 | 24.3       | -26.5 | 24.7       | -27.9 | 25.2       | -29.6 | 25.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.4         | -6.8  | -6.8  | -2.5  | 3.5   | 9.7   | 15.2  | 19.6  | 19.2  | 15.2  | 9.2   | 3.4   | -2.6  | (6)  |  |
|                                                                        | DBStd       | 10.17       | 5.69  | 5.47  | 4.56  | 3.75  | 3.72  | 3.55  | 2.42  | 2.61  | 3.55  | 3.78  | 4.39  | 4.87  | (7)  |  |
|                                                                        | HDD10.0     | 2286        | 521   | 470   | 387   | 198   | 51    | 3     | 0     | 0     | 3     | 61    | 203   | 389   | (8)  |  |
|                                                                        | HDD18.3     | 4463        | 779   | 703   | 646   | 445   | 267   | 104   | 15    | 22    | 104   | 282   | 448   | 648   | (9)  |  |
|                                                                        | CDD10.0     | 990         | 0     | 0     | 0     | 3     | 43    | 159   | 298   | 285   | 159   | 37    | 4     | 0     | (10) |  |
|                                                                        | CDD18.3     | 125         | 0     | 0     | 0     | 0     | 0     | 11    | 54    | 48    | 11    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 472         | 0     | 0     | 0     | 0     | 6     | 48    | 209   | 176   | 33    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 54          | 0     | 0     | 0     | 0     | 0     | 5     | 24    | 21    | 4     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 5.2         | 5.6   | 5.5   | 5.7   | 5.5   | 5.3   | 4.9   | 4.5   | 4.3   | 4.7   | 5.1   | 5.6   | 5.7   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1009        | 67    | 64    | 72    | 78    | 76    | 96    | 73    | 85    | 97    | 106   | 94    | 97    | (15) |  |
|                                                                        | PrecMax     | 1360        | 134   | 118   | 148   | 139   | 187   | 152   | 146   | 161   | 175   | 207   | 160   | 226   | (16) |  |
|                                                                        | PrecMin     | 758         | 9     | 8     | 20    | 40    | 24    | 40    | 32    | 23    | 13    | 28    | 33    | 38    | (17) |  |
|                                                                        | PrecStd     | 160         | 31    | 28    | 34    | 29    | 38    | 29    | 31    | 35    | 47    | 53    | 34    | 48    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 10.0  | 8.1   | 11.5  | 17.4  | 24.0  | 27.0  | 29.2  | 29.2  | 26.5  | 21.1  | 16.9  | 11.4  | (19) |  |
|                                                                        |             | MCWB        | 9.2   | 6.7   | 7.6   | 12.6  | 15.5  | 19.8  | 22.2  | 22.4  | 20.6  | 16.6  | 15.1  | 10.3  | (20) |  |
|                                                                        | 2%          | DB          | 5.9   | 4.8   | 7.4   | 14.1  | 21.0  | 24.6  | 27.0  | 26.7  | 23.9  | 18.3  | 14.4  | 8.5   | (21) |  |
|                                                                        |             | MCWB        | 5.1   | 3.8   | 5.4   | 10.1  | 14.6  | 18.3  | 20.9  | 20.7  | 19.0  | 15.5  | 13.1  | 7.7   | (22) |  |
|                                                                        | 5%          | DB          | 3.3   | 2.7   | 5.2   | 11.6  | 18.3  | 22.9  | 25.6  | 25.2  | 22.1  | 16.7  | 12.2  | 5.9   | (23) |  |
|                                                                        |             | MCWB        | 2.4   | 1.7   | 3.5   | 8.6   | 13.1  | 17.5  | 19.9  | 19.8  | 18.3  | 14.2  | 10.6  | 4.8   | (24) |  |
|                                                                        | 10%         | DB          | 1.2   | 1.1   | 3.4   | 9.3   | 15.9  | 21.1  | 24.2  | 23.9  | 20.5  | 15.0  | 9.9   | 3.8   | (25) |  |
|                                                                        |             | MCWB        | 0.5   | 0.3   | 1.9   | 6.6   | 11.9  | 16.5  | 19.3  | 19.1  | 17.3  | 12.8  | 8.4   | 2.7   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 9.2   | 6.7   | 8.6   | 14.1  | 17.6  | 21.7  | 23.2  | 23.5  | 21.7  | 18.1  | 15.2  | 10.5  | (27) |  |
|                                                                        |             | MCDB        | 10.0  | 7.8   | 10.3  | 16.5  | 21.5  | 25.4  | 27.2  | 27.4  | 25.2  | 19.8  | 16.3  | 11.1  | (28) |  |
|                                                                        | 2%          | WB          | 5.2   | 3.8   | 5.8   | 11.3  | 15.5  | 20.0  | 22.1  | 21.9  | 20.2  | 16.2  | 13.3  | 7.8   | (29) |  |
|                                                                        |             | MCDB        | 5.7   | 4.6   | 7.1   | 13.1  | 19.1  | 22.8  | 25.7  | 25.1  | 22.5  | 17.9  | 14.3  | 8.6   | (30) |  |
|                                                                        | 5%          | WB          | 2.5   | 1.8   | 3.7   | 9.0   | 14.0  | 18.6  | 21.1  | 20.9  | 19.0  | 14.7  | 10.9  | 5.1   | (31) |  |
|                                                                        |             | MCDB        | 3.1   | 2.5   | 4.9   | 11.0  | 17.0  | 21.4  | 24.2  | 23.8  | 21.2  | 16.3  | 12.0  | 5.8   | (32) |  |
|                                                                        | 10%         | WB          | 0.5   | 0.4   | 2.0   | 6.8   | 12.5  | 17.3  | 20.3  | 20.0  | 17.8  | 13.1  | 8.5   | 2.9   | (33) |  |
|                                                                        |             | MCDB        | 1.1   | 1.1   | 3.2   | 9.0   | 15.5  | 20.2  | 23.1  | 22.7  | 20.1  | 14.7  | 9.7   | 3.7   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.7   | 8.0   | 7.0   | 7.2   | 8.5   | 8.4   | 8.0   | 8.0   | 7.9   | 6.9   | 6.0   | 6.2   | (35) |  |
|                                                                        |             | MCDBR       | 10.9  | 8.9   | 8.9   | 11.1  | 12.6  | 11.1  | 10.1  | 9.7   | 9.3   | 8.6   | 8.6   | 8.5   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 11.0  | 8.4   | 7.7   | 7.8   | 7.4   | 6.3   | 5.3   | 5.2   | 5.7   | 6.6   | 8.2   | 8.5   | (37) |  |
|                                                                        |             | MCWBR       | 10.9  | 8.9   | 8.7   | 9.9   | 10.7  | 9.2   | 8.6   | 8.3   | 8.1   | 8.0   | 8.5   | 8.7   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.299 | 0.289 | 0.320 | 0.348 | 0.369 | 0.383 | 0.411 | 0.392 | 0.357 | 0.332 | 0.309 | 0.291 | (40) |  |
|                                                                        |             | taud        | 2.357 | 2.336 | 2.354 | 2.406 | 2.406 | 2.372 | 2.335 | 2.377 | 2.469 | 2.519 | 2.520 | 2.471 | (41) |  |
|                                                                        | Edn at Noon |             | 792   | 887   | 906   | 909   | 897   | 883   | 851   | 854   | 858   | 831   | 787   | 772   | (42) |  |
|                                                                        |             | Edh at Noon | 74    | 91    | 104   | 109   | 114   | 118   | 121   | 112   | 93    | 77    | 63    | 60    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.16  | 1.85  | 3.18  | 4.20  | 5.07  | 5.50  | 5.61  | 4.92  | 3.81  | 2.36  | 1.24  | 0.88  | (44) |  |
|                                                                        | RadStd      |             | 0.09  | 0.23  | 0.28  | 0.40  | 0.48  | 0.41  | 0.38  | 0.31  | 0.34  | 0.24  | 0.15  | 0.08  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (34 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | +0.32 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page